

Query Composition in Composed Image Retrieval: A Survey Summary, Zero-Shot Adaptation, and Formal Formulation for Geo-Conditioned CIR

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Part I — How Prior CIR Work Composes the Query

1.1 The Standard CIR Problem

The Composed Image Retrieval (CIR) task, introduced by Vo et al. (CVPR 2019) with the TIRG method, defines a retrieval problem where the query is inherently multimodal. Rather than searching with a single signal, the user expresses their intent through a **reference image** I_r combined with a **modification text** T_m . The goal is to retrieve a target image I_t from a gallery G such that I_t preserves the relevant visual content of I_r while reflecting the semantic change described by T_m .

Formally, given a dataset of labeled triplets $D = \{(I_r, T_m, I_t)_i\}_{i=1}^N$, the problem is to learn two functions: a **composition function** $f(I_r, T_m)$ that maps the multimodal query to a joint embedding, and a **target embedding function** $h(I_t)$ that maps gallery images into the same space, such that:

$$f(I_r, T_m) \approx h(I_t)$$

Retrieval is then performed by nearest-neighbor search: for a new query (I_r, T_m) , the model returns the top-K gallery images whose $h(I_t)$ embeddings are closest to $f(I_r, T_m)$ under cosine similarity.

The central challenge is that T_m specifies what should *change* while I_r specifies what should *stay the same*. The modification text is inherently differential — it does not describe the target fully but describes a delta. This makes the composition problem fundamentally different from straightforward image-text fusion.

1.2 Supervised CIR: Four-Component Framework

Surveying over 120 papers (Song et al., TOIS 2025; Du et al., TOMM 2025), the architecture of supervised CIR methods consistently decomposes into four stages: feature extraction, image-text fusion, target matching, and (optionally) data augmentation. The field's evolution is largely a story of progressively more powerful fusion strategies.

Feature Extraction

Early work relied on CNN image encoders (ResNet-17 through ResNet-152) paired with sequential text encoders (LSTM, Bi-GRU, GloVe+MLP). From 2022 onward, the field largely converged on **CLIP-based encoders** (ViT-B, ViT-L, RN50x4) for both image and text branches, exploiting their pre-aligned multimodal representation space. More recent methods use BLIP and BLIP-2 encoders for richer cross-modal grounding. The shift to VLP-based encoders is significant: they provide a pre-aligned latent space, reducing the burden on the fusion stage to bridge the visual-semantic gap from scratch.

Image-Text Fusion Strategies

This is the most architecturally diverse component, and the survey identifies five major paradigms:

Explicit Combination-based Fusion is the oldest and most straightforward paradigm. The image and text features are combined through arithmetic or learned gating. The foundational TIRG (2019) computes a transformed-and-residual representation: a gating network learns which parts of the image embedding to suppress, and a residual connection adds the text-transformed features. Later methods like VAL (2020) and DATIR (2021) extended this with hierarchical matching across multiple semantic granularities. The key intuition is that the reference image embedding is treated as a "base vector" and the text embedding applies a directional correction in the embedding space — a form of analogy arithmetic.

Content-and-Style Fusion (CoSMo, 2021) separates the reference image into its structural content (preserved) and its stylistic/attribute features (subject to modification). The text modifier is routed to the style branch while the content branch passes through unchanged. This disentanglement is principled: if the text says "make the collar white," only the style-related dimensions of the embedding should shift.

Neural Network-based Fusion (MLP) treats fusion as a learned feedforward computation. The landmark Combiner (2022) is an MLP that takes concatenated CLIP image and text features and directly predicts the target image embedding. CLIP4CIR (2022) fine-tunes the CLIP backbone end-to-end with a Combiner head. LinCIR (2024) goes further by eliminating the image encoder from training entirely, relying only on text-space operations.

Attention-based Fusion uses cross-attention (e.g., MAAF, ComqueryFormer) or self-attention (e.g., CIRPLANT, FAME-VIL) to allow each modality to dynamically attend to the other. Cross-attention is particularly powerful: the image features become "keys/values" and the text features become "queries," allowing the model to identify which spatial regions of the image are relevant to each word in the modification text. Self-attention jointly encodes the concatenated image patches and text tokens.

Prototype Image Generation-based Fusion takes a generative approach: rather than producing a composed embedding directly, the method synthesizes a "visual proxy" of the target image and uses that as the retrieval query. SynthTripletGAN (2021) was an early version; modern incarnations use latent diffusion (CompoDiff, 2024; CIG, 2025; IP-CIR, 2025) to generate a high-quality visual proxy. This turns retrieval into an image-to-image problem, bypassing the fusion challenge — at the cost of generation time and possible hallucinations.

Target Matching

Once the composed query embedding is computed, target matching determines how the embedding space is structured and how retrieval is scored. Basic metric learning with batch-hard triplet loss is common, but more sophisticated approaches include **image difference alignment** (explicitly training the model to recognize what changed between reference and target pairs), **uncertainty modeling** (CLIP-ProbCR, SDQUR — treating the one-to-many nature of CIR queries as an inherent distributional uncertainty), **negative mining** (constructing hard negatives that are visually similar to the target but semantically incorrect), and **re-ranking** at inference time.

1.3 Zero-Shot CIR (ZS-CIR): Three Paradigms

The high annotation cost of labeled triplets — each requiring human-written modification sentences for reference-target image pairs — has driven substantial interest in zero-shot approaches. The survey classifies ZS-CIR methods into three groups, each making different assumptions about what external information is available.

Paradigm 1 — Textual Inversion-based

The core idea is to map the reference image into the text token embedding space of CLIP, producing a "pseudo-word token" p_I that represents the visual content of I_r as if it were a word. The composed query is then formed by inserting p_I into a text template alongside the modification text: $E_T([p_I, T_m]) \rightarrow q$.

Pic2Word (Saito et al., CVPR 2023) trains a 3-layer MLP projection $P: E_I(I) \rightarrow p_I \in \mathbb{R}^d$ using only image-caption pairs (CC3M), without any labeled triplets. The projection is supervised by a loss that brings $P(E_I(I))$ close to $E_T(\text{caption of } I)$ in CLIP space. At inference, the composed query is $E_T("[p_I] \text{ and } T_m")$.

SEARLE (Baldrati et al., ICCV 2023) takes a similar approach but adds an optimization-based inversion step at test time. It distills this slow optimization into a fast feed-forward network.

LinCIR (Gu et al., CVPR 2024) eliminates the image encoder from training entirely. It uses a Self-Masking Projection that trains on randomly masked CLIP text captions: given a masked sentence, the model predicts the masked span from context, learning a language-only projection that generalizes to images via CLIP's alignment. LinCIR trains in 48 minutes on 8 A100 GPUs yet outperforms prior ZS-CIR methods on CIRCO and CIRR.

The textual inversion paradigm is the most relevant to GeoCIR because it operates entirely in CLIP's shared embedding space — exactly the space that GeoCLIP extends to GPS coordinates.

Paradigm 2 — Pseudo-Triplet-based

Instead of inverting images into text tokens, this paradigm synthesizes approximate training triplets from unlabeled image-text pairs, then trains a standard composition function on those synthetic triplets.

CLIP-CD (2023) generates pseudo-triplets by sampling an image and its caption, treating the image as I_r and the caption as T_m , and using another CLIP-retrieved image as I_t . SDA (2024) uses an LLM to generate modification texts given a reference-target pair. KEDs (2024) uses a dual-stream approach that aligns pseudo-word tokens with textual concepts mined from image-caption pairs.

The key advantage is that pseudo-triplet methods can leverage massive unlabeled corpora. The key risk is that synthetic modification texts may not cover the distribution of real user queries.

Paradigm 3 — Training-Free

The most recent and fastest-growing paradigm avoids any CIR-specific training. These methods decompose the problem into two steps that existing pretrained models can each handle:

1. Use a VLM/MLLM to generate a textual description of the target: $\text{caption}(I_r) + \text{LLM}(T_m) \rightarrow \text{text description of target}$.
2. Use CLIP text-to-image retrieval to find gallery images close to the generated description.

CIReVL (Karthik et al., ICLR 2024) is the canonical training-free method: a captioner (BLIP-2) describes I_r , then GPT-4 fuses the caption with T_m to write a target description, and finally CLIP retrieves by text-to-image similarity. **OSrCIR** (Tang et al., 2025) adds a reflective chain-of-thought step where the model critiques its own generated description and refines it. **TaCIR** (2025) augments the agent with external web search tools to ground the target description.

Training-free methods are powerful but bottlenecked by the quality of the intermediate text generation. They also convert the retrieval problem entirely into the text modality, losing the fine-grained visual grounding of the reference image (e.g., the exact architectural silhouette of Tokyo Skytree).

Part II — Zero-Shot Approaches for GeoCIR

GeoCIR inherits all three ZS-CIR paradigms, but each must be adapted to handle the geographic component of the modifier. Below, each paradigm is analyzed in terms of how it can accommodate a query like ("Tokyo Skytree image", "similar landmark in Germany").

2.1 Textual Inversion Approach for GeoCIR

The textual inversion paradigm is the most naturally compatible with GeoCIR's requirements, because it operates in CLIP's shared embedding space — the same space that GeoCLIP aligns with GPS coordinates.

The method proceeds in three steps. First, the reference image I_r is projected to a pseudo-token $p_I = P(E_I(I_r)) \in \mathbb{R}^d$. Second, the geographic modifier T_g is encoded as $E_T(T_g) \in \mathbb{R}^{512}$ (using CLIP's frozen text encoder, which GeoCLIP has shown aligns with GPS embeddings). Third, the composed query is formed by injecting p_I into a template:

$$q = E_T("a [p_I] located in [T_g]")$$

The key advantage here is that the geographic token in T_g — say, "Germany" — produces a text embedding that, in GeoCLIP's joint space, is geometrically close to the GPS embeddings of German locations. The pseudo-token p_I captures the visual semantics (tall broadcast tower with observation deck), and their joint encoding captures the intent: retrieve an image with those visual properties, but geographically anchored to Germany.

The projection P can be trained using geotagged image-caption pairs from YFCC100M or MP-16, where the captions are augmented with location descriptors. This avoids the need for labeled CIR triplets.

A critical design choice is the template used for composition. Templates like "a $[p_I]$ located in Germany" or "a $[p_I]$ similar to this but in Germany" will produce different embeddings. Template sensitivity is a known limitation of textual inversion methods.

2.2 Pseudo-Triplet Approach for GeoCIR

The pseudo-triplet approach constructs synthetic training examples of the form (I_r, T_g, I_t) without manual annotation. For GeoCIR, these triplets have a clear construction procedure:

1. **Sample a pair of semantically similar landmarks** from different countries: this can be done by finding images in GLDV2 or YFCC100M that share a CLIP image embedding cluster but have GPS coordinates in different countries.
2. **Generate a geographic modifier T_g automatically**: given I_r from Japan and I_t from Germany, the modifier can be templated as "located in Germany" or generated by an LLM as "a similar structure found in Germany."
3. **Train a composition function** on these triplets using the standard triplet loss.

The structural richness of this approach is that the training signal is grounded in real geographic variation. Landmarks that cluster visually across countries (broadcast towers, domed cathedrals, suspension bridges) provide natural positive pairs, while geographically inappropriate retrieval results serve as informative hard negatives.

The challenge is pair construction quality: naively clustering by CLIP similarity may conflate visual similarity with geographic co-occurrence (many iconic towers appear in CLIP training data with their location labels), biasing the model toward memorized associations rather than structural reasoning.

2.3 Training-Free Approach for GeoCIR

The training-free approach adapts most cleanly to the geographic domain because LLMs have rich knowledge about geographic analogy relationships. The pipeline becomes:

1. Caption I_r using BLIP-2: "a tall lattice tower with an observation deck and broadcast antenna."
2. Fuse with T_g using an LLM: "Describe a landmark in Germany that is architecturally similar to a tall lattice tower with an observation deck and broadcast antenna." → "The

Berlin TV Tower (Berliner Fernsehturm), a 368-meter concrete tower with a spherical platform and antenna."

3. Retrieve from gallery using CLIP text-to-image similarity against the generated description.

This approach has a significant advantage for GeoCIR: LLMs are excellent at geographic analogical reasoning ("the Eiffel Tower is to France as the Tokyo Skytree is to Japan"), and can directly name candidate targets. However, it has two limitations. First, it only retrieves well-known landmarks that appear in the LLM's training data, failing for lesser-known structures. Second, it bypasses the image-level visual grounding of I_r , meaning the retrieval relies entirely on the LLM's understanding of visual similarity rather than the actual pixel-level features of the reference.

A hybrid approach — using the LLM to constrain the geographic region but retaining image-level retrieval within that region — may capture the best of both worlds.

Part III — Formal Query Composition for GeoCIR

3.1 Notation and Embedding Spaces

Let E be the shared 512-dimensional embedding space produced by the GeoCLIP framework. Three encoders map into E :

- **Image encoder** $E_I: I \rightarrow E$, where $E_I = \text{CLIP_vision} \circ \text{proj}_{\{h1, h2\}}$. This is the frozen CLIP ViT-L/14 followed by two trainable linear projection layers trained during GeoCLIP's contrastive alignment.
- **GPS (location) encoder** $E_L: (\varphi, \lambda) \rightarrow E$, where (φ, λ) is a latitude-longitude pair. The location encoder applies Equal Earth Projection, M-scale Random Fourier Features with learnable frequency matrix R , and a 4-layer MLP per scale, aggregated into a 512-D embedding.
- **Text encoder** $E_T: T \rightarrow E$. This is the frozen CLIP text encoder followed by the same $\text{proj}_{\{h1, h2\}}$ layers. The text encoder **inherits** the GPS alignment from GeoCLIP: because the image encoder uses a frozen CLIP backbone, and CLIP was trained to align images with text, the projection layers that align images with GPS simultaneously align text with GPS.

The critical alignment property that motivates GeoCIR can be stated as:

Alignment Property. For any image I with associated GPS coordinates (φ, λ) and a text description T of the image's geographic context, the embeddings $E_I(I)$, $E_L(\varphi, \lambda)$, and $E_T(T)$ are geometrically proximate in E : their pairwise cosine similarities are significantly higher than random baseline. This is an emergent property of GeoCLIP's CLIP-backbone and is demonstrated qualitatively in GeoCLIP Section 4.4 (Vivanco Cepeda et al., NeurIPS 2023).

We partition each image embedding conceptually into two latent components: a **visual-semantic component** $v(I)$ that captures the structural and visual properties of the depicted object (architecture, form, material), and a **geographic component** $g(I)$ that captures the location-specific context. These are not explicitly disentangled in GeoCLIP's architecture but are implicitly encoded in the shared embedding space.

3.2 The GeoCIR Task Definition

Definition (GeoCIR Task). Given:

- A reference image I_r depicting a landmark at geographic coordinates (ϕ_r, λ_r)
- A geographic modifier text T_g expressing a target geographic context (e.g., "located in Germany," "similar landmark in Europe")
- A gallery $G = \{I_1, \dots, I_N\}$ of landmark images, each with an associated (unknown to the retriever) GPS coordinate

Retrieve the top-K images $I_{\{(1)\}}, \dots, I_{\{(K)\}}$ from G such that each retrieved image satisfies two simultaneous conditions:

1. **Visual-semantic proximity:** the depicted landmark is architecturally or functionally similar to I_r (same category, similar form, analogous structural role).
2. **Geographic consistency:** the depicted landmark's location is consistent with the target geographic context specified by T_g .

Running example. I_r = photo of Tokyo Skytree (35.710°N, 139.810°E); T_g = "similar landmark located in Germany." Expected top-K results: Berlin TV Tower (52.520°N, 13.409°E), Rhine Tower in Düsseldorf, Hamburg TV Tower. Non-target example: Eiffel Tower (correct visual category, wrong country) or Tokyo Tower (correct country, wrong country) — see geographic specificity discussion in Section 3.5.

The key distinction from standard CIR is that the modifier T_g is **purely geographic** — it does not describe a visual attribute change (color, style, texture) but a *spatial relocation* of the visual concept to a different geographic context. The model must understand that "Germany" in the text encodes a spatial constraint in the GPS embedding subspace of E .

3.3 Query Composition: The Core Formulation

The composed query vector $q \in E$ is produced by a composition function Φ :

$$\boxed{q = \Phi(I_r, T_g)}$$

The retrieval score for a gallery image I_j is:

$$\boxed{\text{score}(I_j) = \cos(q, E_I(I_j)) = (q \cdot E_I(I_j)) / (\|q\| \cdot \|E_I(I_j)\|)}$$

The top-K results are the images with the highest retrieval scores:

$$\boxed{\{I_{\{(1)\}}, \dots, I_{\{(K)\}}\} = \text{argtopK}_{\{I_j \in G\}} \cos(q, E_I(I_j))}$$

We now present four candidate formulations for Φ , ordered from simplest to most principled, each grounded in GeoCLIP's alignment structure.

Formulation A — Linear Composition (Baseline)

The simplest baseline treats the query as a weighted combination of the reference image and the geographic modifier embeddings:

$$\mathbf{q}_A = \alpha \cdot \mathbf{E}_I(I_r) + \beta \cdot \mathbf{E}_T(T_g)$$

where $\alpha, \beta \in \mathbb{R}$ are scalar hyperparameters. This mirrors the "embedding arithmetic" approach used in early CIR work and in Word2Vec analogical reasoning. The implicit assumption is that the visual and geographic information contribute independently additive directions in E .

This formulation requires no training and can be applied directly with any values of α, β (e.g., $\alpha = \beta = 0.5$ for equal weighting). However, it fails to disentangle the geographic component already encoded in $\mathbf{E}_I(I_r)$ — the Tokyo Skytree image embedding already encodes some "Japan-ness," and simply adding $\mathbf{E}_T(\text{"Germany"})$ produces a query that partially represents both locations, which may confuse retrieval.

Formulation B — GPS-Pivot Arithmetic (Proposed Core)

This formulation exploits all three modalities of GeoCLIP's alignment and is the most directly motivated by the GPS-text alignment property. It uses the GPS embedding of the reference location as an explicit pivot:

$$\mathbf{q}_B = \mathbf{E}_I(I_r) - \mathbf{E}_L(\phi_r, \lambda_r) + \mathbf{E}_T(T_g)$$

The intuition is precisely analogous to Word2Vec's "king – man + woman = queen" arithmetic. Here:

- $\mathbf{E}_I(I_r)$ encodes the full visual-geographic embedding of the reference landmark.
- Subtracting $\mathbf{E}_L(\phi_r, \lambda_r)$ removes the geographic component specific to the reference location (Japan/Tokyo), leaving a residual vector that represents the "visual concept without its geographic context."
- Adding $\mathbf{E}_T(T_g)$ injects the target geographic context (Germany), shifting the residual into the target geographic region of E .

This can be rewritten more transparently as:

$$\begin{aligned} \mathbf{q}_B &= [\mathbf{E}_I(I_r) - \mathbf{E}_L(\phi_r, \lambda_r)] + \mathbf{E}_T(T_g) \\ &= \hat{\mathbf{v}}(I_r) + \hat{\mathbf{g}}(T_g) \end{aligned}$$

where $\hat{\mathbf{v}}(I_r)$ is an approximation to the geographic-decontextualized visual embedding, and $\hat{\mathbf{g}}(T_g)$ is the target geographic embedding derived from text.

Requirements for this formulation: the GPS coordinates (ϕ_r, λ_r) of the reference image must be available at query time. In a landmark retrieval setting this is reasonable — the user searching for "similar landmarks to Tokyo Skytree in Germany" presumably knows where Tokyo Skytree is. For a more general setting where GPS is unavailable, the GPS can itself be estimated by GeoCLIP's image-to-GPS retrieval as a pre-processing step.

Formulation C — Geometric Projection (Visual-Geographic Decomposition)

A more principled version of Formulation B that does not require explicit GPS coordinates. This formulation explicitly decomposes the image embedding into visual-semantic and geographic components by projecting onto the geographic subspace of E .

Let $G \subset E$ be the geographic subspace — the manifold of GPS embeddings $E_L(\phi, \lambda)$ for all valid (ϕ, λ) . Let $\text{proj}_G(\cdot)$ denote projection onto G . Then:

$$\begin{aligned} \mathbf{v}_r &= E_I(I_r) - \text{proj}_G(E_I(I_r)) && \text{[visual-semantic residual, } \perp \text{ to geographic subspace]} \\ \mathbf{g}_T &= E_T(T_g) && \text{[target geographic embedding from text]} \\ \mathbf{q}_C &= \mathbf{v}_r + \mathbf{g}_T \end{aligned}$$

This formulation requires characterizing the geographic subspace G . In practice, G can be approximated by sampling a large set of GPS embeddings $\{E_L(\phi_i, \lambda_i)\}$ and computing their principal subspace via PCA, yielding a projection matrix $P_G \in \mathbb{R}^{512 \times k}$. The projection is then $\text{proj}_G(z) = P_G P_G^T z$.

The advantage of Formulation C over B is that it does not require any GPS annotation at inference time. The geographic subspace is pre-computed once from the GPS encoder. The intuition is that the projection onto G extracts "how much GPS information does this image contain," and subtracting it gives a purely visual-semantic vector.

Formulation D — Learned Composition via Textual Inversion

This formulation adapts the textual inversion paradigm (Pic2Word, LinCIR) to the GeoCIR setting. A learned projection network $P: \mathbb{R}^{512} \rightarrow \mathbb{R}^d$ maps the image embedding $E_I(I_r)$ to a pseudo-token p_r in CLIP's text token embedding space. The composed query is then formed by tokenizing a geographic template:

$$\mathbf{q}_D = E_T(["p_r"], T_g) = E_T("a [p_r] located in Germany")$$

where $[p_r]$ is a soft token (a continuous vector in $\mathbb{R}^d$) injected into the CLIP text encoder's input layer at the position of a special token. The text encoder processes the hybrid token sequence (a mix of real word tokens and the continuous pseudo-token) and produces a single 512-D query embedding that jointly encodes the visual reference and the geographic modifier.

The projection network P is trained without labeled CIR triplets, using geotagged image-caption pairs from YFCC100M or MP-16. The training objective is:

$$L_{\text{proj}} = -\log [\exp(\cos(P(E_I(I)), E_T(\text{caption}(I))) / \tau)] / [\sum_j \exp(\cos(P(E_I(I)), E_T(\text{caption}(I_j))) / \tau)]$$

where τ is a temperature parameter and the negative sum runs over a batch of images. This contrastive loss encourages the pseudo-token to be close to the textual description of the image in E , which by CLIP's alignment is also close to geographically consistent GPS embeddings.

The key advantage of Formulation D is that it can be fine-tuned end-to-end and is sensitive to fine-grained visual features that simple arithmetic (Formulations A, B, C) may miss. The trade-off is the need for training and the sensitivity to the template phrasing.

3.4 The Full Retrieval Objective

Collecting the formulations above, the retrieval objective can be stated uniformly. For any composition function $\Phi \in \{\Phi_A, \Phi_B, \Phi_C, \Phi_D\}$:

$$\text{Retrieve } I^* = \operatorname{argmax}_{I_j \in G} \cos(\Phi(I_r, T_g), E_I(I_j))$$

The ground-truth relevance of a gallery image I_j is defined by two conditions that must both be satisfied:

1. **Visual relevance:** the cosine similarity $\cos(E_I(I_j), E_I(I_r))$ exceeds threshold θ_v (i.e., I_j depicts a visually/architecturally similar landmark).
2. **Geographic relevance:** the GPS coordinates (φ_j, λ_j) of I_j fall within the geographic region $R(T_g)$ denoted by the modifier text (e.g., $R(\text{"Germany"}) = \{(\varphi, \lambda) : (\varphi, \lambda) \text{ within Germany's borders}\}$).

These two conditions define a **conjunction** constraint on relevance, which is the novel aspect of GeoCIR compared to standard CIR. In standard CIR, relevance is unidimensional (the target image satisfies the modification). In GeoCIR, relevance is bidimensional: visual similarity AND geographic consistency.

This bidimensional relevance structure motivates using a **factored score**:

$$\text{score}_{\text{factored}}(I_j) = \cos(E_I(I_j), E_I(I_r))^{(1-\gamma)} \cdot \text{geo_score}(I_j, T_g)^\gamma$$

where $\text{geo_score}(I_j, T_g)$ measures geographic consistency (e.g., 1 if $(\varphi_j, \lambda_j) \in R(T_g)$, or a soft score based on $\cos(E_L(\varphi_j, \lambda_j), E_T(T_g))$), and $\gamma \in [0, 1]$ controls the trade-off between visual and geographic components.

Note that Formulations B and C implicitly implement this trade-off through vector arithmetic, while Formulation D learns the trade-off through training. The factored score provides an explicit, interpretable alternative.

3.5 Discussion: Geographic Granularity and Modifier Types

The GeoCIR task naturally accommodates multiple levels of geographic granularity in the

modifier T_g :

Coarse-grained (continental): $T_g = \text{"in Europe"} \rightarrow R(T_g) = \{(\phi, \lambda) : (\phi, \lambda) \text{ in Europe}\}$. Expected results: Eiffel Tower, Berlin TV Tower, Torre de Belém, etc. The model must select among many valid targets, placing emphasis on visual similarity.

Medium-grained (country): $T_g = \text{"in Germany"} \rightarrow R(T_g) = \{(\phi, \lambda) : (\phi, \lambda) \text{ in Germany}\}$. Expected results: Berlin TV Tower, Rhine Tower, Stuttgart TV Tower. More constrained, but visual similarity remains the tiebreaker.

Fine-grained (city): $T_g = \text{"in Berlin"} \rightarrow R(T_g) = \{(\phi, \lambda) : (\phi, \lambda) \text{ in Berlin}\}$. Expected result: Berlin TV Tower (highly constrained both visually and geographically).

The text encoder E_T handles these naturally because CLIP text embeddings encode geographic hierarchy: "Europe" embeddings are centrally placed relative to all European country embeddings, which in turn are centrally placed relative to city embeddings. This hierarchy is an emergent property of the language model's training on geographic text.

There is also a spectrum between **exact geographic specification** ("in Germany") and **relational geographic specification** ("in the country most similar to Japan" or "the European equivalent of Tokyo"). Relational modifiers are richer but harder to ground quantitatively. The initial GeoCIR formulation focuses on exact geographic specifications, leaving relational modifiers to future work.

3.6 Summary of the Four Composition Formulations

Formulation	Expression	GPS Required?	Training?	Key Assumption
A — Linear	$\alpha \cdot E_I(I_r) + \beta \cdot E_T(T_g)$	No	No	Visual and geo contributions add independently
B — GPS Pivot	$E_I(I_r) - E_L(\phi_r, \lambda_r) + E_T(T_g)$	Yes (ref. GPS)	No	GPS arithmetic in E is meaningful
C — Geo Projection	$[E_I - \text{proj}_G(E_I)] + E_T(T_g)$	No	No (offline PCA)	Geographic subspace is separable in E
D — Textual Inversion	$E_T(\text{"[p_r] located in } T_g\text{"})$	No	Yes (P network)	Image maps to CLIP text pseudo-token

Recommended starting point: Formulation B (GPS Pivot) provides the strongest theoretical grounding in GeoCLIP's explicit three-modality alignment and requires no training. It is the most direct exploitation of the GPS-text arithmetic structure that GeoCLIP enables. Formulation

C is the recommended training-free fallback when GPS is unavailable. Formulation D should be explored as the learned counterpart once the arithmetic baselines are evaluated.

References

CIR Surveys:

- Song et al. "A Comprehensive Survey on Composed Image Retrieval." ACM TOIS, 2025. arXiv:2502.18495
- Du et al. "A Survey on Composed Image Retrieval." ACM TOMM, 2025.
- Chen et al. "Composed Image Retrieval: A Survey on Recent Research and Development." Applied Intelligence, 2025.
- Zhang et al. "Composed Multi-modal Retrieval: A Survey of Approaches and Applications." arXiv:2503.01334, 2025.

Foundational CIR Methods:

- Vo et al. "Composing Text and Image for Image Retrieval." CVPR 2019. [TIRG]
- Baldrati et al. "CIRCO: Composed Image Retrieval on Common Objects in Context." ICCV 2023.
- Saito et al. "Pic2Word: Mapping Pictures to Words for Zero-Shot Composed Image Retrieval." CVPR 2023.
- Baldrati et al. "Zero-Shot Composed Image Retrieval with Textual Inversion." ICCV 2023. [SEARLE]
- Gu et al. "Language-Only Training of Zero-Shot Composed Image Retrieval." CVPR 2024. [LinCIR]
- Karthik et al. "Vision-by-Language for Training-Free Compositional Image Retrieval." ICLR 2024. [CIReVL]
- Gu et al. "CompoDiff: Versatile Composed Image Retrieval With Latent Diffusion." TMLR 2024.

GeoCLIP and Geographic Alignment:

- Vivanco Cepeda et al. "GeoCLIP: Clip-Inspired Alignment between Locations and Images for Effective Worldwide Geo-localization." NeurIPS 2023.
- Klemmer et al. "SatCLIP: Global, General-Purpose Location Embeddings with Satellite Imagery." AAAI 2025.
- Haas et al. "Learning Generalized Zero-Shot Learners for Open-Domain Image Geolocalization." ICLR 2023. [StreetCLIP]
- Gurnee & Tegmark. "Language Models Represent Space and Time." ICLR 2024.

- Jia et al. "Towards Interpretable Geo-localization: a Concept-Aware Global Image-GPS Alignment Framework." arXiv:2509.01910, 2025.